

Finite-Window Attachment by Reduced Rational-Frequency Large Sieve

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August 2026

Abstract

TPC-216 proved a complete-period direct-sum envelope for the literal V46 reciprocal emitter but left the physical interval open. This paper attaches that result to the interval $I_x = (x/2, x] \cap \mathbb{Z}$. The common-source kernel is first regrouped exactly by reduced rational frequencies: divisor rows with denominator d contribute to the same frequency a/h through the single coefficient $C_h = \sum_{h|d} \mu(d) \log(d)/d$. Distinct reduced fractions with denominators at most U have circular spacing at least U^{-2} , so the additive large sieve gives a finite-window factor $N + U^2$, where $N = |I_x|$. TPC-215 and TPC-216 then yield

$$N^{-1} \sum_{n \in I_x} |\mathcal{K}(n)|^2 \ll_{\psi} x^{11/32} (\log x)^5.$$

The U^2 term is lower order since $U^2/x = x^{-67/200}$, giving an unnormalized exponent $43/32 + o(1)$. A finite aligned-shell fixture has exact one-point coherent-to-diagonal ratio 2, so the attachment cannot be replaced by free finite-window orthogonality. This is a structural result only: prime-shell reassembly, arithmetic L^2 , the strict $1/400$ budget, and the twin-prime endpoint remain open.

Claim level. PROVED_STRUCTURAL_L1 / FINITE_WINDOW_ATTACHMENT.

The exact regrouping and exponent ledger are proved. The additive large sieve is a standard named input. Finite fixtures are reproducibility checks, not asymptotic arithmetic evidence.

1 Introduction

The transition-native twin-prime route has reached a useful intermediate object. Its physical reciprocal emitter is coupled across a squarefree divisor band, so a divisor-by-divisor orthogonal decomposition is not literal. TPC-213 constructed the common physical source and its cross-divisor Gram [2, 4]. TPC-214 then showed that on a complete period the correct coordinates are reduced rational frequencies. If h is the reduced denominator, all rows with $h \mid d$ share one emitter row B_h and their coefficients add to

$$C_h = \sum_{\substack{d \in \mathcal{D}_x \\ h|d}} c_d, \quad c_d = \frac{\mu(d) \log d}{d}.$$

TPC-215 bounded the resulting cluster coefficient energy by an $O((\log x)^2)$ multiple of the divisor direct-sum energy, and TPC-216 bounded that direct-sum energy by

$$L^{-1}\mathcal{E}_{\text{dir}} \ll_{\psi} x^{11/32}(\log x)^3 [5, 3].$$

Those results still used a complete common period. The next question is whether the same object can be evaluated on the actual interval I_x without silently restoring orthogonality.

The answer is a clean structural attachment. Reduced fractions with denominator at most U form a U^{-2} -separated set modulo one. The additive large sieve therefore controls the physical Gram by $(N + U^2)$ times the reduced-frequency coefficient energy. Since $U^2 = o(x)$, the power exponent is unchanged. The result does not exploit signs of μ , cancellation among primes, or the four-packet polarization. It establishes the finite-window interface and leaves the arithmetic bridge visible.

Our contributions are:

1. an exact finite-window regrouping of the literal common-source kernel;
2. a source-locked large-sieve attachment with normalized exponent $11/32$ and logarithmic cost $(\log x)^5$;
3. an independent finite certificate and a one-point aligned-shell adversary showing why free finite-window orthogonality is invalid.

2 Literal source and inherited complete-period bounds

We retain the V46 parameters

$$H = x^{21/32}, \quad Q = x^{1/3}, \quad Y_0 = \frac{H}{4Q}, \quad U = x^{133/400}, \quad (1)$$

and the prime shell and squarefree band

$$\mathcal{Q}_x = \{q \text{ prime} : Q < q \leq 2Q\}, \quad \mathcal{D}_x = \{d \in \mathbb{Z} : Y_0 < d \leq U, \mu(d)^2 = 1\}. \quad (2)$$

For a bounded profile ψ , define

$$B_d(r) = \sum_{q \in \mathcal{Q}_x} \sum_{0 < |m| \leq \lfloor dq/H \rfloor} \psi\left(\frac{Hm}{dq}\right) \mathbf{1}_{m\bar{q} \equiv r \pmod{d}}. \quad (3)$$

For sufficiently large x ,

$$\frac{H}{4Q} = \frac{1}{4}x^{31/96} \rightarrow \infty, \quad \frac{U}{Q} = x^{-1/1200} \rightarrow 0, \quad (4)$$

so $4Q < H$ and $U < Q$.

The complete-period direct energy from TPC-216 is

$$\mathcal{E}_{\text{dir}} = L \sum_{d \in \mathcal{D}_x} |c_d|^2 \sum_{r \bmod d} |B_d(r)|^2, \quad c_d = \frac{\mu(d) \log d}{d}, \quad (5)$$

with

$$L^{-1}\mathcal{E}_{\text{dir}} \ll_{\psi} \frac{Q^3}{H} (\log U)^3 = x^{11/32} (\log x)^3. \quad (6)$$

TPC-215 supplies the complete-period coefficient comparison

$$\sum_h \sum_{\substack{a \bmod h \\ (a,h)=1}} |C_h B_h(a)|^2 \leq A_x \frac{\mathcal{E}_{\text{dir}}}{L}, \quad A_x = O((\log x)^2), \quad (7)$$

where

$$C_h = \sum_{\substack{d \in \mathcal{D}_x \\ h|d}} c_d. \quad (8)$$

The sum includes only nonzero reduced-frequency rows; the additive zero row vanishes because $q < H$.

3 Exact reduced-frequency representation

Lemma 3.1 (divisor dilation). *If $h \mid d$, $d = kh$, and $(a, h) = 1$, then*

$$B_d(ka) = B_h(a).$$

Proof. The congruence in (3) becomes

$$m\bar{q} \equiv ka \pmod{kh}.$$

It forces $m = kn$ because q is a unit modulo kh . The cutoff becomes $|n| \leq \lfloor hq/H \rfloor$, and the profile argument satisfies $Hm/(dq) = Hn/(hq)$. The summands are therefore identical. $\square$

Theorem 3.2 (exact reduced-frequency regrouping). *Define the common-source kernel*

$$\mathcal{K}(n) = \sum_{d \in \mathcal{D}_x} c_d \sum_{r \bmod d} B_d(r) e(nr/d). \quad (9)$$

Then, for every integer n ,

$$\mathcal{K}(n) = \sum_{h \leq U} \sum_{\substack{a \bmod h \\ (a,h)=1}} C_h B_h(a) e(na/h). \quad (10)$$

Proof. For a residue $r \bmod d$, let $g = (r, d)$ and put $h = d/g$, $a = r/g$. Then $r/d = a/h$ in lowest terms. If $r = 0$, use $h = 1$ and $a = 0$. For nonzero r , the reduced numerator is coprime to h . The map from residues $r \bmod d$ to their reduced fractions partitions every row. Lemma 3.1 identifies the row value in each divisor d with $B_h(a)$, so the coefficients of the same reduced frequency add to C_h . The zero row is absent because $\lfloor q/H \rfloor = 0$ for $q < H$. $\square$

4 Finite-window large-sieve theorem

We use the standard additive large-sieve inequality in the following form [1]. If α_j are separated modulo one by $\delta > 0$, then for every consecutive interval I of N integers,

$$\sum_{n \in I} \left| \sum_j z_j e(n\alpha_j) \right|^2 \leq (N - 1 + \delta^{-1}) \sum_j |z_j|^2. \quad (11)$$

The inequality follows from the dual Hilbert-inequality proof of the additive large sieve; no arithmetic property of the coefficients is required here.

Lemma 4.1 (Farey spacing). *Distinct reduced fractions a/h and a'/h' with $h, h' \leq U$ satisfy*

$$||| (a/h, a'/h') ||| \geq U^{-2},$$

where the distance is taken modulo one.

Proof. The difference, or its complement modulo one, is a nonzero rational with denominator dividing hh' . Its numerator is a nonzero integer, so its absolute value is at least $(hh')^{-1} \geq U^{-2}$. $\square$

Theorem 4.2 (finite-window attachment). *Let $I_x = (x/2, x] \cap \mathbb{Z}$ and $N = |I_x|$. Then*

$$\sum_{n \in I_x} |\mathcal{K}(n)|^2 \ll_\psi (N + U^2) x^{11/32} (\log x)^5. \quad (12)$$

In particular,

$$\frac{1}{N} \sum_{n \in I_x} |\mathcal{K}(n)|^2 \ll_\psi x^{11/32} (\log x)^5, \quad (13)$$

and the unnormalized exponent is $43/32 + o(1)$.

Proof. Insert the exact representation (10) into (11). Lemma 4.1 permits $\delta = U^{-2}$, giving

$$\sum_{n \in I_x} |\mathcal{K}(n)|^2 \leq (N + U^2) \sum_h \sum_{(a,h)=1} |C_h B_h(a)|^2 \quad (14)$$

$$\leq (N + U^2) A_x \frac{\mathcal{E}_{\text{dir}}}{L}. \quad (15)$$

Use $A_x = O((\log x)^2)$ from (7) and (6) to obtain (12). Finally, $U^2/x = x^{-67/200}$ and $N \asymp x$, so division by N gives (13). The unnormalized power is $1 + 11/32 = 43/32$. $\square$

Remark 4.3 (scope). The proof controls the finite-window off-frequency Gram by spacing. It does not use cancellation in $\mu(d)$, cancellation among primes, or the signed four-packet reassembly. The result is consequently structural and does not pay the arithmetic Gate-B ledger.

5 Finite certificates and the crowding obstruction

The reproducible finite fixture uses $Q = \{11, 13, 17\}$, $H = 40$, the full squarefree family in $(2, 35]$, and $\psi(t) = (1 + t^2)^{-2}$. It contains 14 active divisors and 16 reduced denominators. The direct divisor expansion and the reduced-frequency expansion agree pointwise to floating-point rounding in three translated windows of lengths 1, 17, and 60. The finite checks use exact rational row values before the final logarithmic evaluation.

Table 1: Finite-window certificate. The right column is the coarse large-sieve envelope $(N + U^2)S_{\text{cluster}}$.

start	length	window energy	large-sieve envelope
0	1	28.1221	6810.9700
0	17	38.8455	6899.8571
123	60	396.8373	7138.7410

The gap in Table 1 is expected: this release proves a uniform envelope, not a sharp finite fixture estimate. The separate aligned-shell adversary takes

$$d = 5, \quad H = 500, \quad q \in \{101, 131, 151, 181\}, \quad \psi(t) = (1 + t^2)^{-2}.$$

Every row has cutoff one and support $\{1, 4\} \pmod{5}$. At a one-point window, the coherent energy is exactly twice the diagonal row-energy sum. Thus the finite-window Gram is not freely diagonal even in this small admissible model. The statement is scoped: it does not contradict the long-interval large sieve and is not an asymptotic lower bound for the V46 prime shell.

6 Route evaluation and open gates

TPC-217 advances the route from a complete-period envelope to a physical-window attachment. The state is

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TPC217_STRUCTURAL_THRESHOLD_A = PASS,
TPC217_FINITE_WINDOW_ATTACHMENT = PROVED_X_11_OVER_32_LOG_FIVE_NORMALIZED,
TPC217_ARITHMETIC_ADVANCE = NO,
TPC217_PRIME_SHELL_REASSEMBLY = OPEN,
TPC217_FULL_GATE_B = OPEN.
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The next theorem must preserve this finite-window control while restoring the literal prime-only shell, four-packet signs, and the remaining Euler and zero/nonunit interfaces. No fixed-atom credit or strict 1/400 payment is claimed here.

7 Conclusion

The common-source V46 kernel admits an exact reduced-frequency representation on every integer, not only on a complete period. Farey spacing and the additive large sieve then attach the TPC-216 direct-energy envelope to the physical interval with normalized exponent 11/32 and five logarithmic factors. The U^2 spacing term is lower order at the frozen source scales. The one-point aligned-shell fixture shows why this attachment must use a length-dependent large-sieve argument rather than free orthogonality. The remaining work is arithmetic: reassemble the prime shell and four-packet signed scalar without losing the literal Gate-B interface.

References

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